Experiment-1 20/07/26

AIM:
Study of various Network Commands used in Linux / Windows

1. arp-a :-
ARP is short form of address resolution protocol, it will show the IP address of your computer along with IP address and MAC address of your router.

O/P

Internet Address	Physical address
172.16.9.212	7C:57:58:38:da:03
172.16.8.186	7C:57:58:e6:24
172:16.8.249	00:27:0e:13:bf:11
172:16.11.207	7c.59:57:3f:46:f7
17.2: 16.8.131	00.27:0e:13:b4:4)

2. hostname :
This is the simplest of all TCP/IP commands It simply the name of your computer
O/P: localhost.localdomain

3. ipconfig: This command displays detailed configuration information about your TCP/IP connection including Router, gateway, DNS, DHCP and type of Ethernet adapter in your system
O/P: Ethernet adapter Ethernet3.

Connection - Specific DNS Suffix:
Link local IPv6 address ... fe80::cop:4737od
IPv4 address : 192:168:561
Default gateway : 255.255.255.0

Scanned with OKEN Scanner

4. netstat : netstat displays a variety of statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing

Route	Recv Q	Send Q	Local Address	Foreign Addr	State
tcp	0	0	localhost.localdomain:ssh	172.16.8.13:	Establishe
tcp	0	240	localhost.localdomain:ssh	172.16.11.207:59	Est
tcp	0	0	localhost.localdomain:ssh	172.16.8.140.5217	Estab
tcp	0	64	localhost.localdomain:ssh	172.16.8.166.6266	Estab

5. nslookup : name server lookup is a tool used to perform DNS lookup in Linux : It is used to display DNS details such as IP address of particular computer. Mx records for domain.
It operates in two modes : interactive and non-interactive

O/P
www.google.com
Server : 172.16.8.1
Address : 172.16.8.1#53
Name : www.google.com
Address : 142.251.225.119

6. Pathping Pathping is a unique to windows os's basically a combination of the Ping and Tracert commads. Pathping traces the route to the destination address and then launches some.

Ping: Packet Internet Groper command is the best way to test connectivity b/w 2 nodes Ping use ICMP to communicate to other devices

→ Ping www.google.com

Pinging google.com with [142.251.43.142] 32 bytes of data

Reply from 142.251.43.142 bytes=32 time=4ms TTL=118

Reply from 142.251.43.142 bytes=32 time=4ms TTL=118

Ping statistics for 142.251.43.142

Packets: sent=4, Recieved=4, Lost=0 (0% loss)

Approximate round trip times in milli-seconds

Minimum=4ms, Maximum=16ms, Average=7ms

→ Ping Localhost

Pinging DESKTOP-EC436RE [::1] with 32 bytes of data,

Reply from ::1: time<1ms

Reply from ::1: time<1ms

Reply from ::1: time<1ms

Ping statistics for ::1:

Packets: sent=4, Recieved=4, Lost=0

Minimum=0ms, Maximum=0ms, Average=0ms

Linux Networking Commands

1. ip: The ip command is one of the basic commands every administrator will need in daily work sitting up new systems and assigning IPs to troubleshoot existing systems

a) To show the IP address assigned

ip address show

1. link/loopback 00:00:00:00 brd 00:00:00
inet 127.0.0.1/8 scope host to inet 6::1/128 scope host to inet 6::1/128 scope host

2. ifconfig: This command is a staple in many sysadmin tool belt for config and troubleshooting networks. It has since been replaced by ip command

enp 2s0: inet 172.16.9.200

netmask: 255.255.252.0

broadcast: 172:16.11.255

inet6 fe80:c077:e633:8167:72fc

3. ntl : This command combines the functionality of the ping and trace route commands.

a) ntl google.com

Host

1. -gateway
2. 137.97.163.129
3. 172.16.12.122

4. tcpdump: This is designed to capture and display packets

O/P tcpdump -D

1. enp30 (Eth. working)
2. any (Pseudo: device that captures on all interfaces)
3. w/p 370

5. Nbtstat -A

O/P.

NETBIOS's Name Resolution and Registration Statistics

Received by Broadcast = 0

Received by Name Server = 0

Registered by Name Service = 0

6. Pathping -4 172.16.97.153

O/P: Over a max of 30 hops:

0 Lakshi15 [172.16.97.153]

1 Lakshi 15 [172.16.97.153]

Computing statistics for 25 seconds...

Hop	RIT	Source to This Node	Address
1	0ms	0/100 = 0%	0/100 = 0%	172.16.97.153

RESULT:

Successfully executed Linux and Windows Commands